HDOC Day-57

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, **arr[]**, and an integer **k**. Pass the array to a function named **kth_smallest()** using its address. Use pointers to pass the array between the calling function and kth_smallest().

• Arrange the elements of arr[] in ascending order using the bubble sort algorithm.

• Find the element that occupies the kth position in the sorted array.

• Display the kth smallest element.

• Consider duplicate elements as separate elements.

• Display -1 if k is invalid (k <= 0 or k > size).

C Program

```
#include <stdio.h>

void kth_smallest(int *arr, int size, int k)
{
    int i, j, temp;

    if (k <= 0 || k > size)
    {
        printf("-1\n");
        return;
    }

    // Bubble sort in ascending order
    for (i = 0; i < size - 1; i++)
    {
        for (j = 0; j < size - i - 1; j++)
        {
            if (*(arr + j) > *(arr + j + 1))
            {
                temp = *(arr + j);
                *(arr + j) = *(arr + j + 1);
                *(arr + j + 1) = temp;
            }
        }
    }

    printf("kth smallest element: %d\n", *(arr + k - 1));
}

int main()
{
    int size, k, i;

    printf("Enter the size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter the array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", &arr[i]);

    printf("Enter the value of k: ");
    scanf("%d", &k);

    kth_smallest(arr, size, k);
    return 0;
}
```

Test Cases

Test	Array	k	Expected Output
1	7 10 4 3 20 15	3	kth smallest element: 7
2	7 10 4 3 20 15	4	kth smallest element: 10
3	7 10 4 3 20 15	7	-1
4	4 2 2 7 5	3	kth smallest element: 4

Sample Outputs

Example 1

```
Enter the size of array: 6
Enter the array elements: 7 10 4 3 20 15
Enter the value of k: 3
kth smallest element: 7
```

Example 2

```
Enter the size of array: 6
Enter the array elements: 7 10 4 3 20 15
Enter the value of k: 4
kth smallest element: 10
```

Example 3 - Invalid k

```
Enter the size of array: 6
Enter the array elements: 7 10 4 3 20 15
Enter the value of k: 7
-1
```

Example 4 - Duplicate Elements

```
Enter the size of array: 5
Enter the array elements: 4 2 2 7 5
Enter the value of k: 3
kth smallest element: 4
```

Note: In Example 4, the sorted array is 2 2 4 5 7. Duplicate elements are counted separately, so the 3rd smallest element is 4.